

# Zebrafish DEA April 2026

Ishaan Banwait

## Sample Preparation and RNA Extraction/Sequencing (NRC)

- RNA was isolated from zebrafish embryos that had been treated with potential neuroprotective agents in a BMAA chemical model of neurodegeneration
- Libraries were prepared using the Illumina Truseq Stranded RNA kit which uses PolyA enrichment to remove rRNA
- Paired-end sequencing was performed on an Illumina NextSeq 2000 platform
- Performed in two rounds of 3 replicates for each of the 5 conditions
- Total of 30 biological samples (6 reps x 5 conds)
- Each sample underwent paired-end sequencing and generated two files: R1 (forward read) and R2 (backward read) for a total of 60 FASTQ files

## Count Generation (Gurkamal)

*These files were received from [gurkamal@canurta.com](mailto:gurkamal@canurta.com) on January 5th 2026 in a email subject: "Drosophila so far". The files attached are all pertaining to the zebrafish study. The explanation of how files were generated comes from a MS Teams message sent on January 9th 2025, archived at [Kamal Teams Messages](#). "For the zebrafish dataset, I worked with an R environment in the cloud so it was long formatted for R."*

| Step           | Tool             | Input                                                           | Output                                                                          | Notes                                                                                                                                                                                                                                                                  |
|----------------|------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QC             | FastQC + MultiQC | Raw paired-end reads (R1 and R2 aggregated into 30 FASTQ files) | Assessments of read quality, adapter content, duplication, and GC distributions |                                                                                                                                                                                                                                                                        |
| Trimming       | fastp            | Raw reads (30 FASTQ files)                                      | Trimmed reads (30 FASTQ files)                                                  | Trimmed low-quality bases with Phred < 20 (Phred is a measure of how likely a base call is wrong; Phred < 20 means 1% error rate) and reads with <36 base pairs (common minimum for vertebrate genomes; less than this is not alignable)                               |
| Alignment      | HISAT2           | Raw reads (30 FASTQ files)<br>Reference genome                  | 30 SAM/BAM files                                                                | Downloaded the GRCz11 reference genome and the complementary ensemble GTF (annotation). - Used HISAT2 (splice-aware) index to produce SAM (sequence alignment/map) files and then samtools to convert SAM to BAM (binary alignment/map) files for machine readability. |
| Quantification | featureCounts    | 30 BAM files                                                    | raw_counts.txt<br>raw_counts.txt.summary                                        | Reads that mapped ambiguously or with low confidence were automatically ignored.                                                                                                                                                                                       |
| Parsing        | Pandas           | raw_counts.txt                                                  | counts_deseq2.txt<br>(genes x samples)                                          | Removed the extra columns from the featureCounts table and renamed the sample columns from BAM file paths to sample IDs so the count matrix could be used in DESeq2 (genes x samples). This needs to be transposed (samples x genes) for pydeseq2.                     |

## Sample-Level Quality Control

### Sample Screening

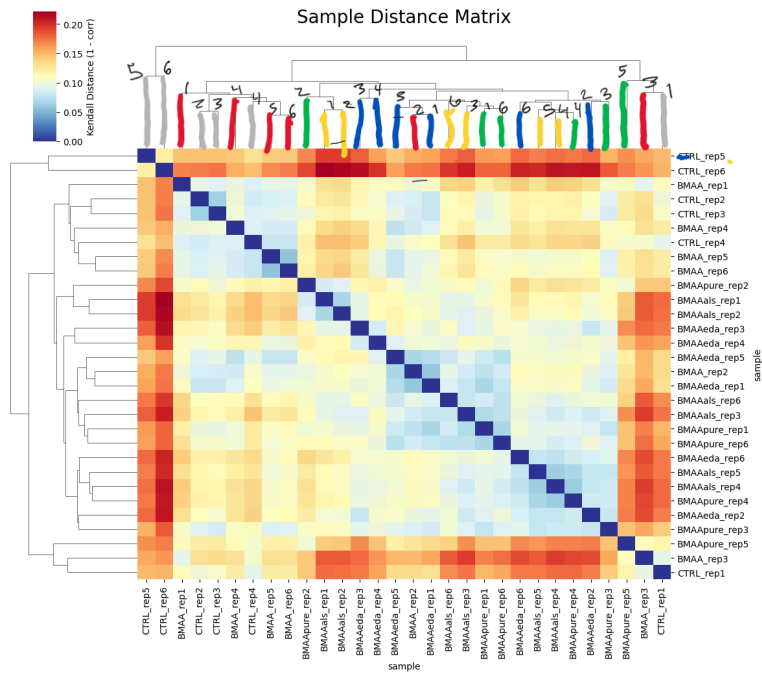
These 4 samples were flagged for violating relative thresholds (beyond 2 MAD from the median in incorrect direction). No absolute thresholds were violated. This is despite the fact that multimapping rates were artificially inflated by using FeatureCounts to count alignments instead of STAR to count reads.

| sample        | relative_thresholds                     | total_reads | library_size | mapping_rate       | n_detected_gei |
|---------------|-----------------------------------------|-------------|--------------|--------------------|----------------|
| BMAA_rep3     | low_mapping_rate                        | 35654113    | 23773774     | 0.6667891022839356 | 25449          |
| BMAAeda_rep6  | low_n_detected_genes                    | 25700188    | 18536313     | 0.721252039090142  | 24226          |
| BMAApure_rep2 | low_mapping_rate,high_multimapping_rate | 30028598    | 20369751     | 0.6783450562693603 | 25421          |
| CTRL_rep2     | high_multimapping_rate                  | 39329094    | 27539680     | 0.7002368272200727 | 25242          |

### Sample Correlation Heatmaps

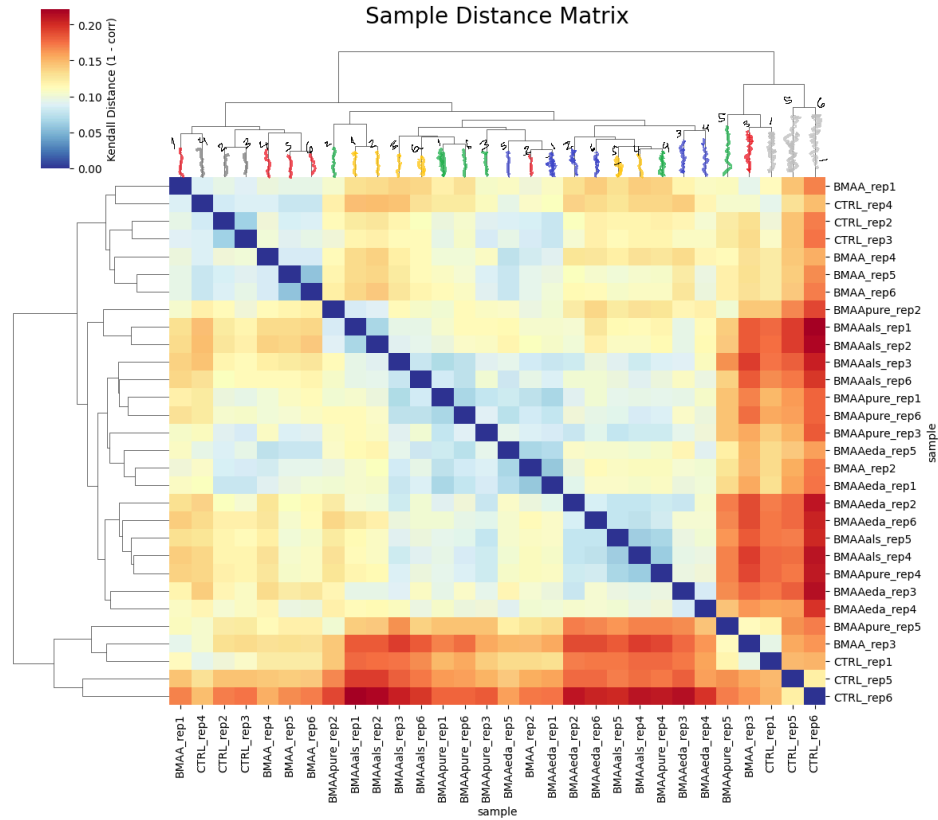
Heatmap of gene expression differences between samples shows healthy controls 5 and 6 in a outgroup on the left (later identified as batch 3). BMAApure\_rep5, BMAA\_rep3, and CTRL\_rep1 are a less distinct outgroup on the right. Then samples separate by controls (healthy and BMAA) and treated, with the exception of BMAA\_rep2 which is a clear outlier. Outliers are:

- BMAApure\_rep5 (right outgroup)
- BMAA\_rep3 (right outgroup)
- CTRL\_rep1 (right outgroup)
- BMAA\_rep2 (in the treated group)



A meta sample correlation heatmap, plotting the differences of differences in gene expression between all samples shows similar clustering, with controls (healthy and BMAA) grouping together and all treated samples in another large group. Understanding (from the PCA) that CTRL\_rep5 and CTRL\_rep6 are from batch effect 3, the following outliers are reproduced:

- BMAApure\_rep5 (right outgroup)
- BMAA\_rep3 (right outgroup)
- CTRL\_rep1 (right outgroup)
- BMAA\_rep2 (in the treated group)



Similar patterns are achieved with a spearman method of correlation. Outliers remain the same.

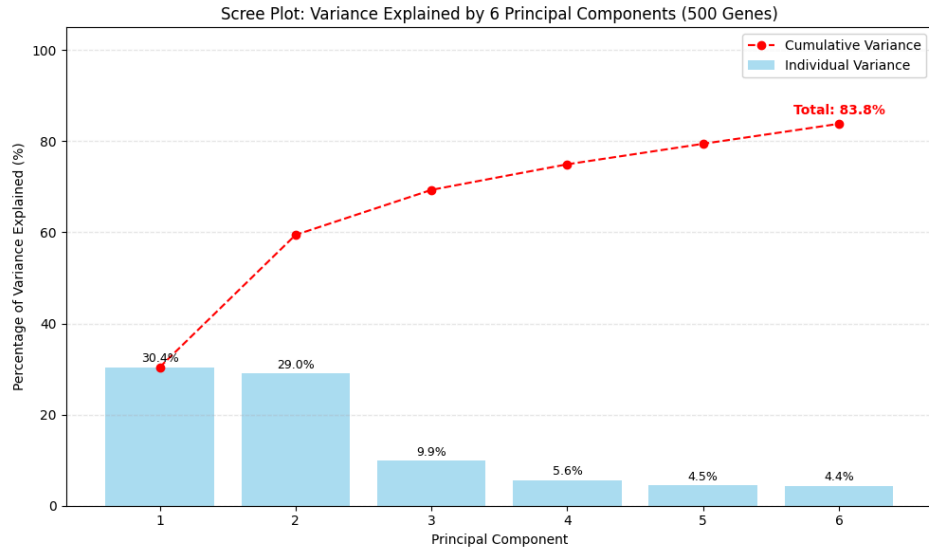
After PCA, the healthy controls 5 and 6 on the right are batch 3, making the other 3 samples in this cluster outliers. Removing these 3 leaves 2 clusters: control (healthy and pathogenic) and treated. Within controls, the lack of distinction between pathogenic and control is probably sex (most likely PC1), which was not recorded. Under this assumption, the pathogenic vs healthy distinction does appear under the parent sex bifurcation.

### Principal Component Analysis

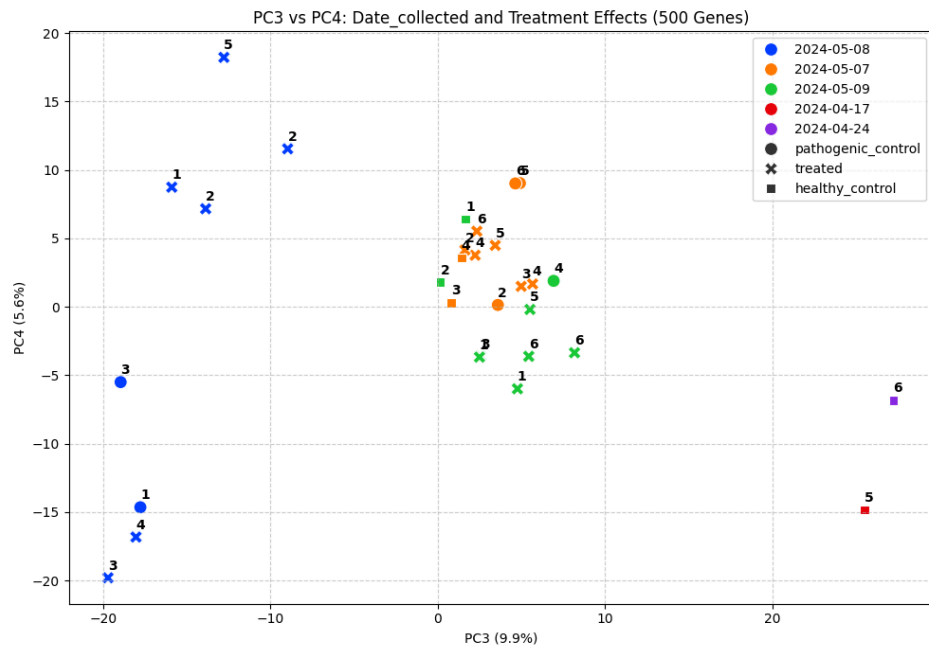
| PC | Variance Explained | Experimental Variable         |
|----|--------------------|-------------------------------|
| 1  | 30.4%              | Probably sex                  |
| 2  | 29.0%              | Treatment vs controls         |
| 3  | 9.9%               | Batch effect (date collected) |

Variation is mostly explained by PC1 and PC2.

Experiment (reps 1-3 vs reps 4-6) doesn't seem to appear on the first 4 PCs, so it doesn't need to be accounted for in design formula.

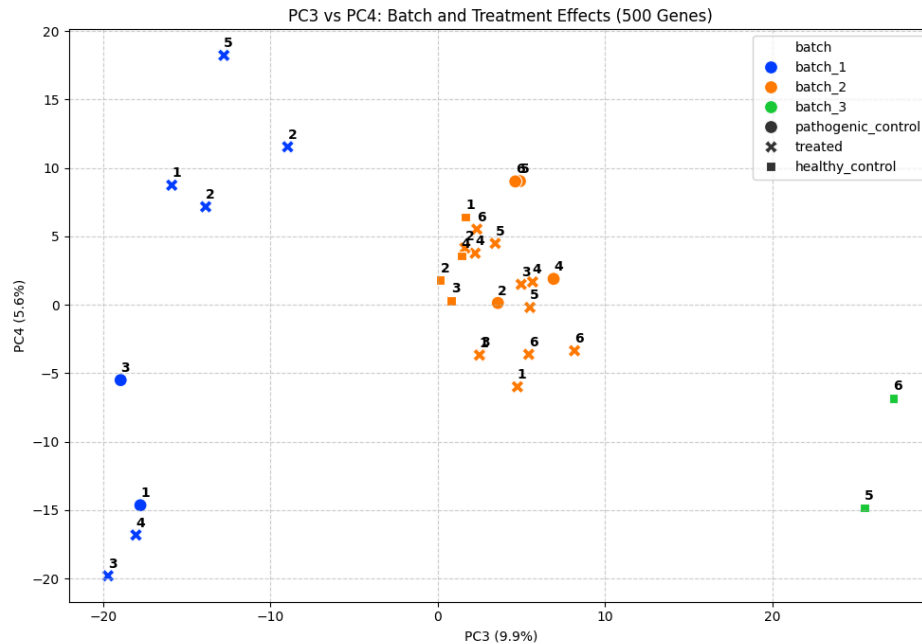


PC3 looks like batch effects from date of specimen collection. Note that batch 3 containing CTRL\_rep5 and CTRL\_rep6 are the most different samples on the heatmap (far right). This exposes BMAApure\_rep5, BMAA\_rep3, and CTRL\_rep1 as potential outliers. These 3 should be considered for removal, since an experimental variable can't be found (and thus statistically controlled for) for these samples.



- Batch 1: May 8th
- Batch 2: May 7th and May 9th
- Batch 3: April 17th and April 24th

Added to the metadata to be controlled for in the design formula.

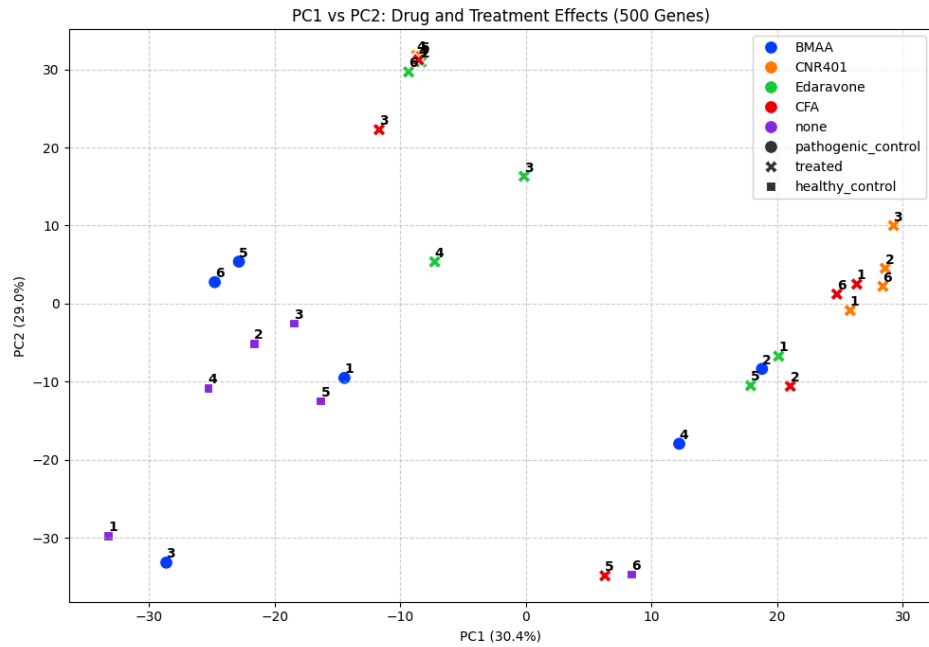


PC1 doesn't look to separate by drug or treatment (vs controls). This could very well be sex which was not recorded, evidenced by the drosophila dataset. An inferred sex column could be added to the metadata to account for this in the design formula. However, the two groups under the controls do not separate by PC1 (inferred sex): BMAA\_rep1 and CTRL\_rep4 (one strata) are near CTRL\_rep2, CTRL\_rep3, BMAA\_rep5, and BMAA\_rep6 (other strata). Based on this, I don't think an artificial sex variable should be applied to the metadata.

BMAA and healthy samples could be clustering together because BMAA (a neurotoxin) acts on the protein level and doesn't cause a transcriptomic impact. This theory is supported by the fact that locomotion was observed to be impaired. With the phenotypic proof, the transcriptomic similarity of healthy and diseased groups isn't as worrying.

PC2 could be treated (vs controls) where drug-treated is higher on PC2. Under this assumption:

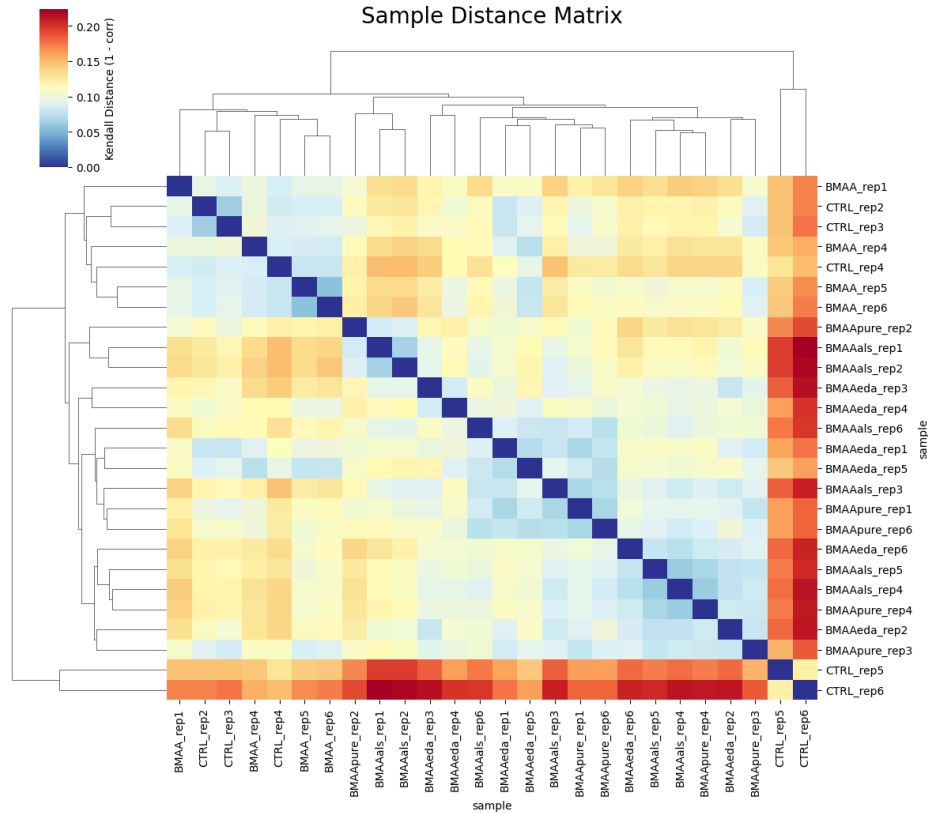
- BMAApure\_rep5 groups with controls
- BMAA\_rep3 and BMAA\_rep2 group with treatments



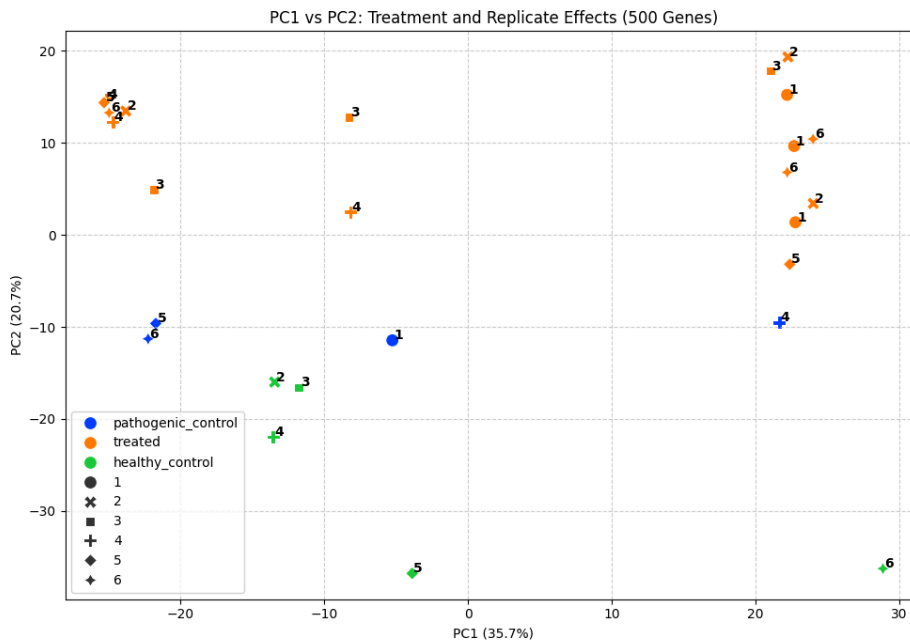
| Sample        | Relative (MAD) violation                | Correlation violation                                 | PCA notes                        | Remove? |
|---------------|-----------------------------------------|-------------------------------------------------------|----------------------------------|---------|
| BMAApure_rep5 |                                         | Once batch 3 is accounted for, exposed as an outlier. | Outlier on PC2.                  | Yes     |
| BMAA_rep3     | low_mapping_rate                        | Once batch 3 is accounted for, exposed as an outlier. | Outlier on PC2.                  | Yes     |
| CTRL_rep1     |                                         | Once batch 3 is accounted for, exposed as an outlier. | Outlier on PC2.                  | Yes     |
| BMAA_rep2     |                                         | Clusters with treatments in heatmap.                  | Clusters with treatments on PC2. | Yes     |
| CTRL_rep2     | high_multimapping_rate                  |                                                       |                                  | No      |
| BMAAeda_rep6  | low_n_detected_genes                    |                                                       |                                  | No      |
| BMAApure_rep2 | low_mapping_rate,high_multimapping_rate |                                                       |                                  | No      |
| CTRL_rep5     |                                         | Furthest outgroup.                                    | Batch 3                          | No      |
| CTRL_rep6     |                                         | Furthest outgroup.                                    | Batch 3. Outlier on PC2.         | No      |

### Post-Filtering Sample Correlation and PCA

The heatmap looks much better, with a batch 3 outgroup and then a cluster for controls and treated samples.



PCA also looks better, with sex on PC1 and treatment on PC2.



Because of the similarity between both positive and negative controls, it may be a good idea to group them together into a single control group to increase sample size and statistical power. BMAA vs healthy can still be controlled for as a cofactor. Right now there is 6 CNR401, 5 CFA, 6 Edaravone, 4 BMAA, and 5 healthy.

## Sample QC Conclusion

- 4 samples highlighted in red will be removed from DEA.
- Batch effects will be adjusted for in the design formula
- Sex cannot be accounted for, which is a limitation of this study. However, differentially expressed genes/gene sets that are caused by sex should be easy to manually pick out and exclude from discussion. To preserve power and avoid mislabeling samples, an artificial sex label will not be applied.

## Differential Expression Analysis

### Gene and Sample Filtering

There were 26 samples after dropping the 4 outliers above. Genes with <10 counts were filtered out, decreasing the number of genes from 32520 to 26436.

Metadata dataframe that entered the `DeseqDataSet`.

| sample        | drug      | treatment          | replicate | date_collected | batch   |
|---------------|-----------|--------------------|-----------|----------------|---------|
| BMAA_rep1     | BMAA      | pathogenic_control | 1         | 2024-05-08     | batch_1 |
| BMAA_rep4     | BMAA      | pathogenic_control | 4         | 2024-05-09     | batch_2 |
| BMAA_rep5     | BMAA      | pathogenic_control | 5         | 2024-05-07     | batch_2 |
| BMAA_rep6     | BMAA      | pathogenic_control | 6         | 2024-05-07     | batch_2 |
| BMAAals_rep1  | CNR401    | treated            | 1         | 2024-05-08     | batch_1 |
| BMAAals_rep2  | CNR401    | treated            | 2         | 2024-05-08     | batch_1 |
| BMAAals_rep3  | CNR401    | treated            | 3         | 2024-05-09     | batch_2 |
| BMAAals_rep4  | CNR401    | treated            | 4         | 2024-05-07     | batch_2 |
| BMAAals_rep5  | CNR401    | treated            | 5         | 2024-05-07     | batch_2 |
| BMAAals_rep6  | CNR401    | treated            | 6         | 2024-05-09     | batch_2 |
| BMAAeda_rep1  | Edaravone | treated            | 1         | 2024-05-09     | batch_2 |
| BMAAeda_rep2  | Edaravone | treated            | 2         | 2024-05-07     | batch_2 |
| BMAAeda_rep3  | Edaravone | treated            | 3         | 2024-05-08     | batch_1 |
| BMAAeda_rep4  | Edaravone | treated            | 4         | 2024-05-08     | batch_1 |
| BMAAeda_rep5  | Edaravone | treated            | 5         | 2024-05-09     | batch_2 |
| BMAAeda_rep6  | Edaravone | treated            | 6         | 2024-05-07     | batch_2 |
| BMAApure_rep1 | CFA       | treated            | 1         | 2024-05-09     | batch_2 |
| BMAApure_rep2 | CFA       | treated            | 2         | 2024-05-08     | batch_1 |
| BMAApure_rep3 | CFA       | treated            | 3         | 2024-05-07     | batch_2 |
| BMAApure_rep4 | CFA       | treated            | 4         | 2024-05-07     | batch_2 |
| BMAApure_rep6 | CFA       | treated            | 6         | 2024-05-09     | batch_2 |
| CTRL_rep2     | none      | healthy_control    | 2         | 2024-05-09     | batch_2 |
| CTRL_rep3     | none      | healthy_control    | 3         | 2024-05-07     | batch_2 |
| CTRL_rep4     | none      | healthy_control    | 4         | 2024-05-07     | batch_2 |
| CTRL_rep5     | none      | healthy_control    | 5         | 2024-04-17     | batch_3 |
| CTRL_rep6     | none      | healthy_control    | 6         | 2024-04-24     | batch_3 |

### Statistical Design

- A single `DeseqDataSet` was fit, so that there was one global GLM per gene for all 26 samples. This allows for more statistical power when estimating gene-wide dispersions since more replicates are available, helping with controlling the noise in this messy dataset.



- A reference-means model was used with drug=none and batch=batch\_1 as the reference levels (intercept;  $\beta_0$ ). The other two batches (2 and 3) were each given a coefficient referenced to batch\_1 ( $\beta_2, \beta_3$ ). The four drugs (BMAA and 3 treatments) were each given a coefficient referenced to no drug ( $\beta_3, \beta_4, \beta_5, \beta_6$ ). Contrasts can be extracted relative to either negative (n=4) or positive (n=5) controls. There is little difference between the controls. It was decided to reference to negative controls to match the structure of the Drosophila experiment.

*The design matrix used in this analysis.*

| sample        | Intercept ( $\beta_0$ ) | batch[T.batch_2] ( $\beta_1$ ) | batch[T.batch_3] ( $\beta_2$ ) | drug[T.BMAA] ( $\beta_3$ ) | drug[T.CNR401] ( $\beta_4$ ) | drug[T.Edaravone] ( $\beta_5$ ) | drug ( $\beta_6$ ) |
|---------------|-------------------------|--------------------------------|--------------------------------|----------------------------|------------------------------|---------------------------------|--------------------|
| BMAA_rep1     | 1.0                     | 0                              | 0                              | 1                          | 0                            | 0                               | 0                  |
| BMAA_rep4     | 1.0                     | 1                              | 0                              | 1                          | 0                            | 0                               | 0                  |
| BMAA_rep5     | 1.0                     | 1                              | 0                              | 1                          | 0                            | 0                               | 0                  |
| BMAA_rep6     | 1.0                     | 1                              | 0                              | 1                          | 0                            | 0                               | 0                  |
| BMAAals_rep1  | 1.0                     | 0                              | 0                              | 0                          | 1                            | 0                               | 0                  |
| BMAAals_rep2  | 1.0                     | 0                              | 0                              | 0                          | 1                            | 0                               | 0                  |
| BMAAals_rep3  | 1.0                     | 1                              | 0                              | 0                          | 1                            | 0                               | 0                  |
| BMAAals_rep4  | 1.0                     | 1                              | 0                              | 0                          | 1                            | 0                               | 0                  |
| BMAAals_rep5  | 1.0                     | 1                              | 0                              | 0                          | 1                            | 0                               | 0                  |
| BMAAals_rep6  | 1.0                     | 1                              | 0                              | 0                          | 1                            | 0                               | 0                  |
| BMAAeda_rep1  | 1.0                     | 1                              | 0                              | 0                          | 0                            | 1                               | 0                  |
| BMAAeda_rep2  | 1.0                     | 1                              | 0                              | 0                          | 0                            | 1                               | 0                  |
| BMAAeda_rep3  | 1.0                     | 0                              | 0                              | 0                          | 0                            | 1                               | 0                  |
| BMAAeda_rep4  | 1.0                     | 0                              | 0                              | 0                          | 0                            | 1                               | 0                  |
| BMAAeda_rep5  | 1.0                     | 1                              | 0                              | 0                          | 0                            | 1                               | 0                  |
| BMAAeda_rep6  | 1.0                     | 1                              | 0                              | 0                          | 0                            | 1                               | 0                  |
| BMAApure_rep1 | 1.0                     | 1                              | 0                              | 0                          | 0                            | 0                               | 1                  |
| BMAApure_rep2 | 1.0                     | 0                              | 0                              | 0                          | 0                            | 0                               | 1                  |
| BMAApure_rep3 | 1.0                     | 1                              | 0                              | 0                          | 0                            | 0                               | 1                  |
| BMAApure_rep4 | 1.0                     | 1                              | 0                              | 0                          | 0                            | 0                               | 1                  |
| BMAApure_rep6 | 1.0                     | 1                              | 0                              | 0                          | 0                            | 0                               | 1                  |
| CTRL_rep2     | 1.0                     | 1                              | 0                              | 0                          | 0                            | 0                               | 0                  |
| CTRL_rep3     | 1.0                     | 1                              | 0                              | 0                          | 0                            | 0                               | 0                  |
| CTRL_rep4     | 1.0                     | 1                              | 0                              | 0                          | 0                            | 0                               | 0                  |
| CTRL_rep5     | 1.0                     | 0                              | 1                              | 0                          | 0                            | 0                               | 0                  |
| CTRL_rep6     | 1.0                     | 0                              | 1                              | 0                          | 0                            | 0                               | 0                  |

*The contrasts used in this analysis, of format ['variable\_of\_interest', 'tested\_level', 'ref\_level'].*

["drug", "CNR401", "BMAA"]

["drug", "CFA", "BMAA"]

["drug", "Edaravone", "BMAA"]